

Fruit bat vocalization
translation and behavior
observation automation
powered by A.I.

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ABSTRACT:

- Keepers in zoos need to order food for bats in advance but currently have no way of predicting how much they’ll eat, resulting in money wasted.
 - More data on activity is needed to find patterns in consumption.
 - We developed a hybrid CNN-LSTM model to recognize context and species of vocalizations and automate continuous activity data collection.
- METHODOLOGY:
- **Training:** isolate single vocalizations and group clips based on spectrogram analysis. Label each group with zookeeper. Build A.I. model using Keras
 - **Model application:** Build a graphical user interface (GUI) for keepers to generate vocal activity graph and log data into spreadsheet.
 - **Analysis:** match vocal activity with appetite the next day for strength of correlation.

AI MODEL ARCHITECTURE:

- Hybrid Convolutional Neural Network–Long Short Term Memory (CNN-LSTM) architecture.
- Input layer: Spectrogram features (79, 120).
 - 1st Convolutional block: 64 filters to learn local spatial frequency patterns (39, 64).
 - 2nd Convolutional block: 128 filters learning combinations of Block 1 patterns (19, 128).
 - LSTM: Learns how frequency spatial patterns change over duration of vocalization (128,).
 - Classification/dense layers: fully-connected layer with 128 neurons, identifying class-discriminative features – combinations of LSTM outputs used in distinguishing behaviors.
 - Output layer: Uses softmax activation function to identify probability of each context.

RESULTS:

- There is a moderate correlation between the straw want-food calls and the food consumption the next day.
 - The more hungry, the more the bats eat.
 - Can use trendline on scatter plot to predict food consumption the next day based on vocal activity.

CONCLUSION:

- Vocalization data over 14 days from Oregon Zoo were purposefully collected for this study.
- Our neural network model could identify the context of bat vocalizations & species with 99% accuracy.
- New overnight recordings were analyzed using the model to generate activity frequency graphs.
 - Offer new insight on bat’s overnight behaviors.
- Statistical correlations between overnight vocal activities and appetite the following day were successfully established.
- GUI was developed for zookeepers to collect data on a daily basis for more accurate appetite predictions.
- An educational zoo cart and website were built to educate the guests about bats and their vocalizations. Our vocalization model was published on Github (batAI).



Educational bat
vocalization website



Oregon Zoo

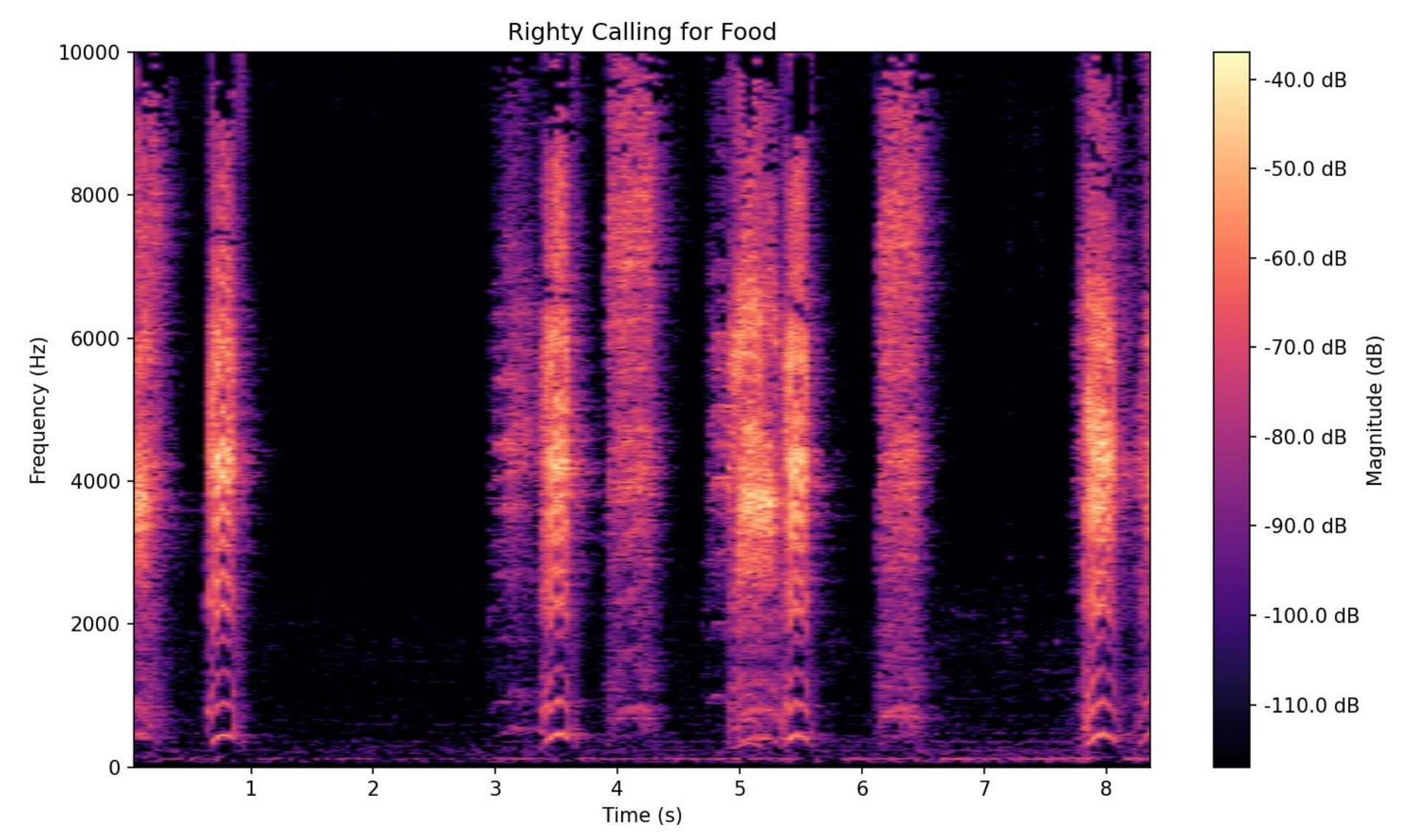


Figure 1: A spectrogram of a Straw-Colored bat named Righty calling for food.



Figure 2: our GUI, which maps the overnight behavior, logs it into a spreadsheet, and generates a visual frequency chart.

Date	Total Hours	Rods Fighting	Rods Talking	Straws Fighting	Straws Fighting Talking	Straws Talking	Straws Want Food	Total Calls Per Hour	Adjusted consumption
1/28/26	9.3	0	0	30.3	0.2	1.1	0.2	31.8	26
1/30/26	16.9	0	0	15.1	0.1	0.4	0.5	16.1	27.5
1/31/26	18	0	0	14.6	0.1	1.2	1.8	17.6	55
2/1/26	4.9	0	0	0	0	0	0	0	65
2/2/26	4.8	0.6	0.2	91.7	4.4	18.5	32.3	147.7	65
2/6/26	9.3	0.1	0	2.2	0	0	4.1	6.3	90
2/7/26	9.3	0	0	25.1	0	0	7.2	32.3	100
2/8/26	18.6	0.3	0	29.4	0.2	2.4	5.2	37.4	52
2/21/26	13.2	0.1	0	37	0.2	3.4	2.1	42	45.5
2/20/26	18.6	0.1	0	35.6	0.5	3.4	2.7	42.4	30
2/19/26	18.6	0	0	22.4	0.1	1.2	2.3	26	30
2/23/26	18.6	0.2	0	32.7	0.5	2.1	2.7	38.2	65
2/24/26	18.6	0.2	0	41.8	0.8	8.2	5.2	56.1	65
2/25/26	14.8	0	0	38.2	1.4	4.9	2.4	47	90
2/28/26	27.9	0	0	33.8	0.2	0.7	3.9	38.7	70
2/27/26	18.6	0.1	0.1	34.8	0.6	2.8	1.2	39.6	75
3/1/26	15.1	0	0.1	59.8	0.3	3.2	1.7	64.9	39

Figure 3: an example of our spreadsheet with the data for each of the calls.

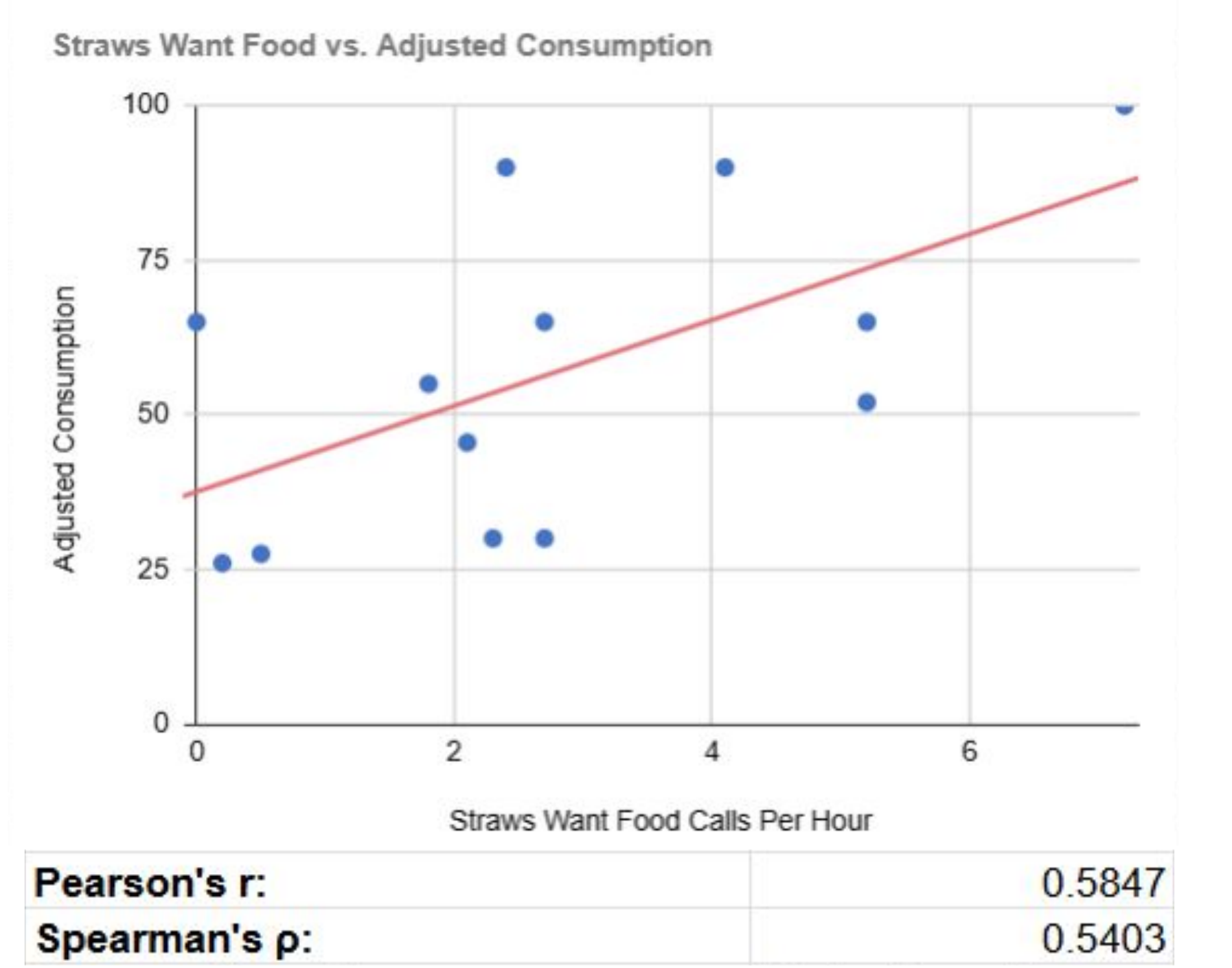


Figure 4: The correlation between Straws want food calls and adjusted food consumption the next day.